Hyperspectral Image Denoising via Transformed Tensor Nuclear Norm and Nonconvex Regularization in Spatial–Spectral Gradient Domains

Duo Qiu, Tao Li, and Xiongjun Zhang

***Abstract*—Hyspectral images (HSIs) are inevitably degraded by a mixture of different types of noise, such as Gaussian noise, impulse noise, stripes, and deadlines. Traditional methods exploit the global low-rankness and local smoothness by using two separate regularization terms for HSI denoising, which may be highly affected by the trade-off parameters between the two regularizers. In this article, we propose to utilize the three-directional tensor correlated total variation (3DTCTV) to explore the global low-rankness and local smoothness of HSIs simultaneously. More specifically, the three-directional transformed tensor nuclear norm (TNN) is employed to the gradient domain of the underlying HSIs along the spatial and spectral dimensions. And the tensor ℓ_1 norm is used to characterize the sparsity of the noisy tensor. Moreover, a nonconvex approach is proposed to characterize the low-rankness of the underlying HSIs and enhance the sparsity of noisy tensor further for HSI denoising, where a family of nonconvex functions are employed onto the singular values of the three-directional gradient tensors along spatial–spectral dimensions in the transformed domain and each entry of the sparse tensor, respectively. An alternating direction method of multipliers is developed to solve the resulting models, and its convergence for the nonconvex model is established under some conditions. Extensive numerical experiments on simulated and real HSI datasets demonstrate that the proposed methods outperform various state-of-the-art methods both visually and quantitatively.**

***Index Terms*—3-D tensor correlation total variation, gradient domain, hyperspectral image (HSI) denoising, nonconvex regularization, transformed tensor nuclear norm (TNN).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spatial and spectral information and have been widely used in unmixing, classification, object detection, material identification, and anomaly detection [2], [47], [74]. However, in the acquisition and transmission process, HSIs inevitably suffer from different kinds of noises due to the atmospheric inference and signal response [53], [74]. The degradation of HSIs caused by several types of noises (such as Gaussian noise, impulse noise, stripes, and deadlines) can severely influence the performance of subsequently high-level applications of HSIs. Therefore, it is an important preprocessing step to remove the mixed noise for improving HSI quality in further applications.

There exist a huge number of works for HSI denoising in the literature, which are generally divided into three categories: model-driven methods, data-driven methods, and model–data-driven methods [74]. Leveraging on the deep learning framework, data-driven approaches for HSI denoising can learn the trainable parameters from external datasets and labels, which include supervised and unsupervised network architectures. For supervised networks, there exist plenty of work in the literature, such as 2-D convolutional neural network (CNN) [33], [67], 3-D CNN [30], [44], and hybrid networks [3], [78]. Furthermore, some unsupervised methods for HSI denoising were also proposed and studied, see [32], [73] and references therein. However, these methods need a large number of clean HSIs for training in practice, which is hard to obtain in most imaging environments.

For the model–data-driven methods, they can overcome the disadvantage of the model-driven- and data-driven-based methods, and preserve the superiority of the two kinds of methods efficiently for HSI denoising. The model–data-driven techniques exploit their representational capacity to learn the nonlinear relationship between clean and noisy HSIs, and also leverage the priors of HSIs, see [9], [10], [26], [27], [38], [43], [58], [69], [73] and references therein. For example, Lin et al. [27] proposed an HSI denoising framework for removing the mixed Gaussian and impulse noise, in which the denoising problem is modeled as a CNN method with constrained non-negative matrix factorization. Zhang et al. [69] proposed a novel end-to-end low-rank spatial–spectral network for the removal of the hybrid noise

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in HSIs, which integrates the low-rank physical property into a deep CNN. By integrating model-driven with data-driven strategy, Zhang et al. [73] proposed a self-supervised HSI denoising method, which simultaneously cooperates the spectral low-rankness prior and deep spatial prior. Li et al. [26] proposed a CNN driven by the structural knowledge of noise for HSI denoising, where the knowledge optimization-driven module employed the deep unrolling method to unfold a total variation (TV) algorithm by using structural characteristics of noise. Shi et al. [43] proposed a double subspace deep prior model by using two deep neural networks, where the TV regularizer was added to constrain the generation of the subspace factor neural network. Chen et al. [10] presented a plug-and-play deep low-rank decomposition (DLD) model into the optimization framework, which incorporated traditional prior knowledge into the DLD model. Peng et al. [38] proposed a specific optimization framework for learning a representative coefficient image denoiser under the low-rank decomposition framework for HSI denoising. Zhuang et al. [79] presented to utilize the low-rank property of HSIs in the spectral dimension and leverage the spatial correlation of images through CNN for HSI denoising. Ye et al. [65] proposed an iterative low-rank network based on a progressive refinement process and spectral low-rankness prior for effective HSI denoising, which integrated the strengths of model-driven and data-driven approaches by embedding a rank minimization module within a U-Net architecture. Chen et al. [9] proposed a novel self-supervised HSI reconstruction framework by using the traditional model-driven low-rank tensor prior and the implicit deep spatial–spectral prior. Nevertheless, the generalization ability of the model–data-driven-based methods is not stable and relatively weak for different kinds of noises for HSI denoising [74].

The model-driven-based methods utilize the intrinsic structure of HSIs for mixed noise removing, where HSIs contain abundant spatial–spectral characteristics. In general, piecewise smoothness, nonlocal self-similarity, and low-rankness of HSIs are explored for HSI denoising. Some widely used methods by using piecewise smoothness prior were proposed to explore the local details for HSI denoising via TV, which contains the spatial and spectral TV. For example, Aggarwal and Majumdar [1] proposed a spectral–spatial TV (SSTV) method to explore the spectral–spatial smoothness of HSIs for HSI denoising. Moreover, other TV-based methods were also proposed and studied in the literature, such as spectral–spatial adaptive TV (SSATV) [66], enhanced 3-D TV [39], and representative coefficient TV [37].

There exist many similar and correlated local patterns for HSIs, where the nonlocal self-similarity methods can be exploited via small local structures. Early works regarding nonlocal similarity were based on the spatial nonlocal similar prior to remove noise band-by-band, such as nonlocal means filter-based method [11], dictionary learning-based method [36], BM3D [12], and weighted nuclear norm minimization [16]. However, these works ignore the correlation of HSIs along the spectral dimension. Furthermore, some methods based on spatial and spectral nonlocal self-similarity were proposed and studied for HSI denoising, such as BM4D [34], which leveraged on both local voxel correlation within cubes and nonlocal correlation between spatially corresponding voxels across cubes.

Low-rankness plays a vital role for HSI denoising, which reflects the global correlation for HSIs along the spatial and spectral dimensions. By unfolding each band into a vector and stacking these vectors into a matrix for HSIs, some low-rank matrix-based methods for HSI denoising were proposed and studied in the literature. For example, Zhang et al. [70] proposed a low-rank matrix recovery method via low-rank and sparse decomposition to remove the mixed noise of HSIs. In order to explore the noise intensity in different bands, He et al. [19] proposed a noise-adjusted iterative low-rank matrix approximation method for HSI denoising. By leveraging on a better approximation of the rank function of a matrix, Chen et al. [5] proposed a nonconvex low-rank matrix approximation method via γ-norm for HSI denoising, and Ye et al. [64] proposed a smooth rank approximation by using different nonconvex functions in the low-rank component for HSI denoising, which can explore the low-rankness of a matrix better than the nuclear norm-based methods.

On the other hand, the low-rank matrix-based methods combined with other priors were also proposed and studied for HSI denoising in the literature, such as nonlocal prior, TV prior, and sparse prior. For example, Xue et al. [61] proposed a joint spectral and spatial low-rank-regularized method for HSI denoising, which incorporated the low-rankness-based nonlocal similarity into sparse representation. Yang et al. [63] proposed a Moreau-enhanced TV-regularized method for HSI denoising, which used the low-rank prior, local smoothness, and nonlocal similarity of HSIs. Peng et al. [37] proposed a fast model-based approach via a representative coefficient TV regularizer to simultaneously characterize the low-rank and local smooth properties. More work regarding low-rank matrix-based methods for HSI denoising can be found in [20], [48], [59] and references therein. Nevertheless, the low-rank matrix-based methods ignore the original structure of HSIs, which destroy the intrinsic correlation of HSIs via the unfolding approaches.

Since an HSI is a cube, that is, a third-order tensor, low-rank tensor-based methods are powerful for HSI denoising, which can exploit the intrinsic correlation of HSIs better [55]. However, different from the rank of a matrix, the definition of tensor rank is more complex due to its non-uniqueness. The widely used tensor ranks are CANDECOMP/PARAFAC (CP) rank [22], Tucker rank [51], tensor tubal rank and multirank [24], tensor train rank [35], and tensor ring rank [76]. For example, Guo et al. [17] proposed a rank-one tensor CP decomposition method with the use of an eigenvalue intensity sorting and reconstruction technique for HSI denoising. By using the nonlocal self-similarity technique, Xue et al. [60] proposed a nonlocal tensor CP decomposition method via exploiting the global correlation across spectrum for HSI denoising. However, computing the CP rank of a tensor is generally NP-hard [21]. For HSI denoising based on Tucker decomposition, Wang et al. [55] proposed to combine tensor Tucker decomposition with SSTV for HSI denoising, which can remove mixed noise efficiently. However, the above work

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needs to choose a suitable Tucker rank for good denoising performance, which may be difficult. By using nonlocal similarity and high spectral correlation, Zhang et al. [72] proposed a novel method for HSI denoising by combining nonlocal low-rank tensor decomposition and TV regularization. Chen et al. [6] proposed a group sparsity regularization of the spatial difference images for HSI restoration, which used the tensor Tucker decomposition and the group sparsity regularization for the clean HSIs. Chang et al. [4] proposed a weighted low-rank tensor recovery model by treating the singular values differently for HSI restoration, which used Tucker decomposition in the similar cubes. He et al. [18] proposed a unified paradigm by integrating the spatial nonlocal similarity with the global spectral low-rank property for HSI restoration, which jointly learned and updated the orthogonal basis matrix and reduced image. However, the above work did not remove the sparse noise with high noise levels efficiently. Recently, Su et al. [49] proposed a subspace representation method along the spectral dimension with weighted multidirectional low-rank tensor regularization and nonlocal self-similarity. More works regarding HSI denoising based on tensor Tucker decomposition can be found in [4], [25], [50] and references therein. Based on tensor train or tensor ring decomposition, Wang et al. [54] proposed a tensor-based method by modeling two intrinsic HSI priors, namely nonlocal low-rank tensor train and 3-D weighted TV for spectral compressive imaging. Chen et al. [7] proposed a nonlocal tensor ring decomposition method for HSI denoising. Su et al. [49] proposed a unidirectional spatial and spectral smoothed tensor ring decomposition for HSI denoising and destriping, where the tensor ring decomposition was introduced to explore the global spatial–spectral correlation of HSIs. Zhang et al. [71] proposed to reconcile sparse and low tensor ring rank priors in the learnable transformed domain for HSI denoising, where the tensor ring decomposition was used to explore the low-rankness of HSIs in the transformed domain. However, the previous tensor ring-based methods should give the tensor ring rank in advance, which may be difficulty to tune in general.

Based on tensor singular value decomposition (SVD) [24], Fan et al. [13] proposed a tensor nuclear norm (TNN) method to approximate the multirank of a tensor for HSI denoising, where the TNN was defined as the sum of nuclear norms of all frontal slices in the transformed domain. Furthermore, Fan et al. [14] proposed an SSTV-regularized low-rank tensor factorization method to remove mixed noise in HSIs, where the TNN regularization and rank constraint were both imposed in the model. However, the above two works only discussed the nuclear norms of spectral dimension of HSIs, which may be challenged since the HSIs are low rank along both spatial and spectral dimensions. Then Zheng et al. [77] proposed a three-directional TNN-based method for HSI denoising and its nonconvex relaxation, which can characterize the low-rankness better than TNN since both spatial and spectral information of HSIs were utilized. Instead of using the convex and nonconvex surrogates to approximate the tubal rank of three modes, Liu et al. [31] proposed a fibered rank-constrained tensor restoration framework with an embedded plug-and-play-based regularization for HSI denoising. Moreover,

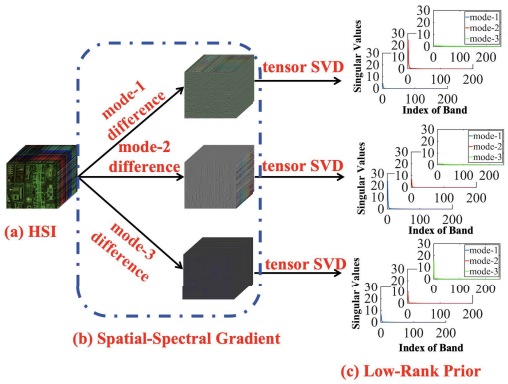


Fig. 1. Illustrations of low-rankness of the gradient tensors along different modes for the Washington DC Mall dataset. (a) Original HSI. (b) Gradient tensors along different modes. (c) Corresponding distributions of singular values of the gradient tensor along different modes by using DCT.

Liu et al. [28] proposed a gradient subspace-regularized TNN-based method by applying mode-3 low-rank decomposition of the gradients of HSIs to obtain their representation coefficient tensors for HSI denoising. Recently, Xue et al. [62] proposed a tensor convolution-like low-rank dictionary method for high-dimensional image representation, which can be applied for removing the sparse noise of HSIs. Here the definition of the tensor convolution-like low-rank dictionary is based on the TNN of two smaller tensors, where the underlying tensor is factorized into the tensor–tensor product of the two small tensors. However, the TNN-based methods just used the nuclear norm of all frontal slices of a tensor or tensor–tensor product in the Fourier domain. There is a limitation in the use of discrete Fourier transform, where periodicity is assumed. Moreover, the discrete Fourier transform may not result in a low-rank tensor [45], while a low-rank requirement is preferred for HSI denoising. Recently, Song et al. [45] defined tensor–tensor product based on any unitary transform and proposed a transformed TNN (TTNN) for robust tensor completion, which can explore the low-rankness of a tensor better under suitable unitary transformations. More applications regarding TTNN can be found in [40], [41], [42], [46], [57], and [75]. However, the TTNN only explores the low-rankness of a tensor along the third-dimension, while many tensors are both low-rank and sparse in the transformed domain. Recently, Wang et al. [52] proposed a tensor correlation TV (TCTV) for low-rank tensor recovery, which can explore the low-rankness and sparsity in the transformed domain using a unique regularizer. More applications regarding TCTV can be found in [23], [29] and references therein. However, the TCTV only utilized the TTNN on the gradient tensor along the third-dimension, which may be challenged since the gradient tensor of HSIs is low-rank along three modes, see Fig. 1. Moreover, the TTNN is just the convex envelope of the sum of transformed multirank of a tensor, which may be suboptimal.

In this article, we propose a three-directional TCTV (3DTCTV) for removing the mixed noises of HSIs, where the 3DTCTV can characterize the low-rankness and smoothness

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by using the unique TCTV term along different modes and can explore the two priors simultaneously. More specifically, the three-directional TTNN is employed to the three-gradient tensors of the underlying HSIs along different modes in 3DTCTV. Compared with TCTV, the 3DTCTV can explore the low-rankness and smoothness better since the three-directional TTNN is used on the gradient tensors. Besides, the tensor ℓ_1 norm is utilized to explore the sparsity of the sparse noise. Furthermore, a nonconvex method based on 3DTCTV is proposed to enhance the low-rankness and smoothness of HSIs, and sparsity of sparse noise for HSI denoising, where a family of nonconvex functions are employed onto the singular values of the gradient tensors along different modes in the transformed domain and each element of the sparse noise, respectively. An alternating direction method of multipliers (ADMM)-based algorithm is designed to solve the resulting models, and its convergence for the nonconvex model is established under some conditions. Extensive numerical experiments on simulated and real datasets demonstrate the effectiveness of the proposed methods over other state-of-the-art methods for HSI denoising.

The main contributions of this article are summarized as follows.

1) A 3DTCTV method and its nonconvex formulation are proposed for removing the mixed noises of HSIs, which can explore the low-rankness and smoothness of HSIs using a unique regularizer.
2) An ADMM-based algorithm is presented to solve the resulting models, and the convergence of ADMM for the nonconvex model is established.
3) Extensive experiments are conducted in both simulated and real data conditions to illustrate the superiority of the proposed models for HSI denoising.

The remaining parts of this article are organized as follows. In Section II, some preliminaries regarding generalized tensor SVD under any unitary transformations are reviewed. In Section III, a 3DTCTV model and its nonconvex relaxation are proposed for the mixture noise removal of HSIs. In Section IV, an ADMM-based algorithm with convergence guarantee is designed to solve the resulting nonconvex model. Extensive numerical experiments on simulated and real data for HSI denoising are conducted to demonstrate the superiority of the proposed methods in Section V. Finally, the conclusions are drawn in Section VI. The convergence analysis of the ADMM algorithm and more experiments are deferred to Supplementary Material.

II. Preliminaries

Throughout this article, scalars, vectors, matrices, and tensors are denoted by lowercase letters (e.g., a), bold lowercase letters (e.g., $\mathbf{a}$), bold uppercase letters (e.g., $\mathbf{A}$), and calligraphic letters (e.g., $\mathcal{A}$), respectively. Diag($\mathbf{a}$) denotes a diagonal matrix with the ith diagonal element being a_i, where a_i denotes the ith element of $\mathbf{a}$. For any matrix $\mathbf{A}$, $\mathbf{A}^T$ denotes the conjugate transpose of $\mathbf{A}$. For a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$, the (i,j,k)th element of $\mathcal{A}$ is represented by $\mathcal{A}_{ijk}$. The mode-1, mode-2, and mode-3 slices are called horizontal, lateral, and frontal slices, respectively. The ith mode-k slice of $\mathcal{A}$ is denoted by $\mathbf{A}_k^{(i)}$. The inner product of two tensors is defined as $\langle \mathcal{X}, \mathcal{Y}\rangle = \sum_{i=1}^{n_1}\sum_{j=1}^{n_2}\sum_{k=1}^{n_3} \mathcal{X}_{ijk}^* \mathcal{Y}_{ijk}$, where $\cdot^*$ denotes the conjugate operator. The tensor Frobenius norm of $\mathcal{A}$ is defined as $\|\mathcal{A}\|_F = (\langle \mathcal{A}, \mathcal{A}\rangle)^{1/2}$. The tensor ℓ_1 norm of a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ is defined as $\|\mathcal{A}\|_1 = \sum_{i,j,k} |\mathcal{X}_{ijk}|$. The n-mode (matrix) product of a tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times\cdots\times n_d}$ with a matrix $\mathbf{U} \in \mathbb{R}^{p\times n_k}$ is denoted by $\mathcal{X} \times_k \mathbf{U} \in \mathbb{R}^{n_1\times\cdots n_{k-1}\times p\times n_{k+1}\cdots\times n_d}$, and its element is defined as $(\mathcal{X}\times_k \mathbf{U})_{i_1\ldots,i_{k-1}ji_{k+1},\ldots,i_d} = \sum_{i_k=1}^{n_k} \mathcal{X}_{i_1,\ldots,i_k,\ldots,i_d}\mathbf{U}_{ji_k}$. The nuclear norm of a matrix, denoted by $\|\cdot\|_*$, is defined as the sum of all singular values of the matrix. The proximal mapping of a function $g : \mathbb{R} \to \mathbb{R}$ is defined as $\mathrm{Prox}_g(y) := \arg\min_x\{g(x) + 1/2\|x-y\|^2\}$.

A. Generalized Tensor SVD via Unitary Transformations

In this section, we generalize the tensor–tensor product under any unitary transform [45] to three directions for third-order tensors. For any third-order tensor $\mathcal{A} \in \mathbb{C}^{n_1\times n_2\times n_3}$ and a unitary matrix $\mathbf{U}_{n_k} \in \mathbb{R}^{n_k\times n_k}$, we define $\widehat{\mathcal{A}}_k := \mathcal{A}\times_k \mathbf{U}_{n_k}, k = 1,2,3$. The mode-$k$ block diagonalization operation of $\mathcal{A}$ and its inverse operation are defined as follows:

$$\mathrm{bdiag}\,(\mathcal{A},k) = \begin{pmatrix} \mathbf{A}_k^{(1)} & & \\ & \ddots & \\ & & \mathbf{A}_k^{(n_k)} \end{pmatrix}$$

and $\mathrm{bfold}(\mathrm{bdiag}(\mathcal{A},k),k) = \mathcal{A}$.

Based on [77, Definition 1], we can generalize the definition of mode-k tensor product from Fourier transform to any unitary transform.

Definition 1 (Mode-k Product): For two tensors $\mathcal{A}$ and $\mathcal{B}$, the mode-k tensor product under a unitary transform matrix $\mathbf{U}$ is defined as $\mathcal{A}\diamond_k\mathcal{B} = \mathrm{bfold}(\mathrm{bdiag}(\widehat{\mathcal{A}}_k,k)\cdot\mathrm{bdiag}(\widehat{\mathcal{B}}_k,k),k)\times_k\mathbf{U}_{n_k}^T$.

Similar to [77, Sec. III], we can define mode-k conjugate transpose, mode-k identify tensor, diagonal tensor, orthogonal tensor, and mode-k tensor SVD via any unitary transform.

Definition 2 (Mode-k Conjugate Transpose): For a third-order tensor $\mathcal{A} \in \mathbb{C}^{n_1\times n_2\times n_3}$, the mode-$k$ conjugate transpose of $\mathcal{A}$ via a unitary matrix $\mathbf{U}_{n_k} \in \mathbb{C}^{n_k\times n_k}$ is defined as $\mathcal{A}^{T_k} = \mathrm{bfold}((\mathrm{bdiag}(\widehat{\mathcal{A}}_k,k))^T)\times_k\mathbf{U}_{n_k}^T$.

Definition 3 (Mode-k Identify Tensor, Diagonal Tensor, and Orthogonal Tensor): The model-k identity tensor, denoted as $\mathcal{I}_k$, is defined as $\mathcal{I}_k := \mathcal{T}_k\times_k\mathbf{U}_{n_k}^T$, where all mode-$k$ slices of $\mathcal{T}_k$ are identity matrices. A third-order tensor is called a mode-k diagonal tensor if each mode-k slice is a diagonal matrix. A third-order tensor $\mathcal{Q}$ is mode-k orthogonal if $\mathcal{Q}\diamond_k\mathcal{Q}^{T_k} = \mathcal{Q}^{T_k}\diamond_k\mathcal{Q} = \mathcal{I}_k$.

Definition 4 (Mode-k Tensor SVD): For a tensor $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$, the mode-$k$ tensor SVD is given as follows: $\mathcal{A} = \mathcal{U}_k\diamond_k\mathcal{S}_k\diamond_k\mathcal{V}_k^{T_k}$, where $\mathcal{U}_k$ and $\mathcal{V}_k$ are the mode-k orthogonal tensors, and $\mathcal{S}_k$ is the mode-k diagonal tensor.

Now we define the three-directional TTNN based on [45, Definition 7] and [77, Definition 9].

Definition 5: For a tensor $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and unitary matrix $\mathbf{U} \in \mathbb{R}^{n_k\times n_k}$, the three-directional transformed TNN (3DTTNN) is defined as follows: $\|\mathcal{A}\|_{\mathrm{3DTTNN}} = \sum_{k=1}^3 \alpha_k\|\mathrm{bdiag}(\widehat{\mathcal{A}}_k,k)\|_*$, where $\alpha_k \geq 0$ and $\sum_{k=1}^3\alpha_k = 1$.

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TABLE I
DIFFERENT NONCONVEX FUNCTION $g_1(x)$

Function	$g_1(x)$
SCAD $(a>0, b>1)$	$\begin{cases} a\lvert x\rvert, & \text{if } \lvert x\rvert < a, \\ \frac{2ab\lvert x\rvert - x^2 - a^2}{2(b-1)}, & \text{if } a \le \lvert x\rvert < ab, \\ \frac{(b+1)a^2}{2}, & \text{if } \lvert x\rvert \ge ab. \end{cases}$
MCP $(c, \eta > 0)$	$\begin{cases} c\lvert x\rvert - \frac{1}{2\eta}x^2, & \text{if } \lvert x\rvert \le c\eta, \\ \frac{c^2\eta}{2}, & \text{if } \lvert x\rvert > c\eta. \end{cases}$
ℓ_q norm	$\lvert x\rvert^q,\ 0 < q < 1$
logarithmic	$\log\left(1 + \frac{\lvert x\rvert}{\theta}\right),\ \theta > 0$

Definition 6: For a tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$ and unitary matrix $\mathbf{U} \in \mathbb{R}^{n_k \times n_k}$, the 3-D nonconvex penalty (3DNP) based on TTNN is defined as follows:

$$\text{3DNP}(\mathcal{A}) := \sum_{k=1}^{3} \alpha_k \text{NP}_k(\mathcal{A}) \tag{1}$$

where $\text{NP}_k(\mathcal{A}) := \sum_t g_1(\sigma_t(\text{bdiag}(\widehat{\mathcal{A}}_k, k)))$ with $\sigma_t(\cdot)$ being the tth largest singular value of the matrix $\text{bdiag}(\widehat{\mathcal{A}}_k, k)$, $\alpha_k \ge 0$, $\sum_{k=1}^{3} \alpha_k = 1$, and $g_1 : \mathbb{R} \to \mathbb{R}$ is a nonconvex function satisfying Assumption 1.

Assumption 1: g is concave, nondecreasing on $(0, \infty)$, and symmetric regarding y-axis with $g(0) = 0$.

In Definition 6, the nonconvex function g_1 is employed onto the singular values of the block matrix $\text{bdiag}(\widehat{\mathcal{A}}_k, k)$, $k = 1, 2, 3$. Moreover, a lot of nonconvex functions in the literature satisfy Assumption 1, such as smoothly clipped absolute deviation (SCAD) [15], minmax concave penalty (MCP) function [68], ℓ_q norm $0 < q < 1$, and logarithmic function (log) [40], [41], [42], which are summarized in Table I.

The 3DTCTV is a generalization of the TCTV in [52, Definition 9], which considers three-directional TTNN of the gradient tensor. The definition of 3DTCTV is given as follows.

Definition 7: For any tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$, the 3DTCTV is defined as $\|\mathcal{A}\|_{\text{3DTCTV}} = 1/3 \sum_{i=1}^{3} \|\mathcal{G}_i\|_{\text{3DTTNN}}$, where $\mathcal{G}_i$ is the gradient tensor of $\mathcal{A}$ along the ith mode defined by $\mathcal{G}_i = \mathcal{A} \times_i \mathbf{D}_{n_i}$ with $\mathbf{D}_{n_i} \in \mathbb{R}^{n_i \times n_i}$ being a row circulant matrix of $(-1, 1, 0, \ldots, 0)$.

On the one hand, the 3DTCTV norm constrains the 3DTTNN metric of the gradient tensors $\mathcal{G}_i$ along three directions, which is capable of promoting the low-rankness of the three-gradient tensors along different directions. In fact, the transformed tubal rank of $\mathcal{G}_i$ is between $r - 1$ and r, where r is the transformed tubal rank of $\mathcal{A}$ [52]. Therefore, the 3DTCTV can encode the low-rankness of the original tensor. On the other hand, the 3DTCTV norm and the TV norm are consistent to control the trend of changes of the underlying tensor $\mathcal{A}$ in local regions since they are all certain measure constraints on the gradient tensors. In fact, the two norms tend to be smaller when the original data becomes smoother and takes the minimum value of zero only if data is absolutely flat. This indicates that the 3DTCTV has potential to constraint the smoothness prior structure of the underlying tensor to some extent, which is similar to the TV norm.

III. PROPOSED MODEL FOR HSI DENOISING

A. Problem Statement

For a clean HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it may be corrupted by mixed noises including Gaussian noise and sparse noise (such as impulse noise, stripes, and deadlines) simultaneously. The HSIs degraded procedure can be formulated as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N}$$

where $\mathcal{S}$ and $\mathcal{N}$ denote sparse noise and Gaussian noise, respectively. The aim of HSI denoising is to restore the clean HSI $\mathcal{X}$ from the noisy image $\mathcal{Y}$.

B. 3DTCTV and Nonconvex Relaxation Models for HSI Denoising

For the clean HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we propose to employ the TCTV-based method to characterize the low-rankness and smoothness of HSIs simultaneously. Since the spectral correlation of HSIs, the HSI (third-order tensor) is low-rank. Recently, the TCTV is proposed to characterize the low-rankness and smoothness of the underlying tensor for tensor recovery [52]. However, the TCTV only considers the low-rankness and smoothness of the third-dimension of the gradient tensors, which may be challenged. In Fig. 1, we show the distributions of singular values of the gradient tensors of the Washington DC Mall dataset along different modes, where the discrete cosine transform (DCT) is used in the transformed tensor SVD. It can be observed that the singular values of the gradient tensor along different modes decrease greatly and only few singular values are large, which shows the approximate low-rankness of the gradient tensor along different modes. Therefore, in order to explore the low-rankness and smoothness of HSIs better, we propose to employ the 3DTCTV to characterize the clean HSIs. The main superiority of 3DTCTV is that the three-directional TTNN of each gradient tensor is used for HSIs, which can explore the low-rankness and smoothness of HSIs efficiently. Moreover, the tensor ℓ_1 norm is used to measure the sparsity of the sparse noise. Then the resulting model (called 3DTCTV) for HSI denoising is given as follows:

$$\begin{aligned} \min_{\mathcal{X},\mathcal{N},\mathcal{S}} \quad & \|\mathcal{X}\|_{\text{3DTCTV}} + \lambda_1 \|\mathcal{N}\|_F^2 + \lambda_2 \|\mathcal{S}\|_1 \\ \text{s.t.} \quad & \mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \end{aligned} \tag{2}$$

where $\lambda_1, \lambda_2 > 0$ are given constants.

Inspired by the nonconvex relaxation for sparse and low-rank optimization problems, we propose a 3-D nonconvex penalty for HSI denoising (called 3DNP) as follows:

$$\begin{aligned} \min_{\mathcal{X},\mathcal{N},\mathcal{S}} \quad & G_1(\mathcal{X}) + \lambda_1 \|\mathcal{N}\|_F^2 + \lambda_2 G_2(\mathcal{S}) \\ \text{s.t.} \quad & \mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \end{aligned} \tag{3}$$

where $G_1(\mathcal{X}) := 1/3 \sum_{i=1}^{3} \text{3DNP}(\mathcal{G}_i)$ with $\text{3DNP}(\cdot)$ being defined in (1), $\mathcal{G}_i$ is the gradient tensor of $\mathcal{X}$ along the ith mode defined by $\mathcal{G}_i = \mathcal{X} \times_i \mathbf{D}_{n_i}$ with $\mathbf{D}_{n_i} \in \mathbb{R}^{n_i \times n_i}$ being a row circulant matrix of $(-1, 1, 0, \ldots, 0)$, and $G_2(\mathcal{S}) := \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \sum_{k=1}^{n_3} g_2(\mathcal{S}_{ijk})$. Here g_2 also satisfies Assumption 1 and can be the nonconvex functions in Table I. The main

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aim of $G_2(\mathcal{S})$ is used to get a sparser solution compared with the tensor ℓ_1 norm, which will be helpful for HSI denoising. In fact, the ℓ_1 norm is used onto each element of the sparse tensor for the sparse component in model (2). However, the ℓ_1 norm is just suboptimal and obtains biased estimators in statistical learning [15]. Then some nonconvex relaxation approaches with statistical guarantee are proposed to approximate the ℓ_0 norm better, such as SCAD and MCP [15], [68]. Compared with the tensor ℓ_1 norm in model (2), the nonconvex term $G_2(\mathcal{S})$ in model (3) can explore the sparsity of the sparse noise better, which can benefit from HSI denoising conversely.

In model (3), a family of nonconvex functions g_1 and g_2 are employed onto the singular values of the three-gradient tensors of $\mathcal{X}$ along different modes in the transformed domain and each entry of the sparse tensor, respectively. The superiority of g_1 for the low-rank component is similar to that of g_2 for the sparse component. In fact, g_1 is used to the singular values of the three-gradient tensors, which can explore the low-rankness and smoothness of the clean HSIs better compared with 3DTCTV.

Remark 1: The main idea of model (3) is that a family of nonconvex functions are used to approximate the ℓ_0 norm instead of the ℓ_1 norm for the low-rank and smooth component, and sparse component, respectively, where the nonconvex functions satisfying Assumption 1 can approximate the ℓ_0 norm better by the concavity of these functions [15], [40], [41], [68]. In model (3), the nonconvex functions in Table I can be used for both g_1 and g_2. In particular, we will compare the denoising performance of different nonconvex functions of 3DNP by numerical experiments in Section V-C.3.

Remark 2: We give the comparisons between our models (2) and (3), and the low-rank tensor factorization model in [62]. The model in [62] utilizes the tensor convolution-like low-rank dictionary to explore the low-rankness of the underlying tensor, where the underlying tensor is factorized into the tensor–tensor product of two smaller tensors and takes the TNN onto the small factor tensors. However, we use the 3DTCTV and its nonconvex relaxation to explore the low-rankness and smoothness of HSIs in models (2) and (3). Moreover, our models in (2) and (3) are used to remove mixed noises of HSIs, such as Gaussian noise, impulse noise, stripes, and deadlines, while the model in [62] is just used to remove sparse noise for the problem of tensor principal component analysis.

IV. ADMM-Based Algorithm

Since model (2) is just the special case of model (3), where $G_1(\mathcal{X})$ and $G_2(\mathcal{S})$ are replaced by $\|\mathcal{X}\|_{\text{3DTCTV}}$ and $\|\mathcal{S}\|_1$, respectively, we only consider to solve model (3) in detail.

By introducing auxiliary variables $\mathcal{G}_{ki}$, we can reformulate model (3) as follows:

$$\begin{aligned}&\min_{\mathcal{X},\mathcal{G}_{ki},\mathcal{S},\mathcal{N}} \ \frac{1}{3}\sum_{i=1}^{3}\sum_{k=1}^{3}\alpha_k \text{NP}_k\left(\mathcal{G}_{ki}\right)+\lambda_1\|\mathcal{N}\|_F^2+\lambda_2 G_2\left(\mathcal{S}\right)\\ &\text{s.t.} \quad \mathcal{Y}=\mathcal{X}+\mathcal{N}+\mathcal{S}, \nabla_i\left(\mathcal{X}\right)=\mathcal{G}_{ki}, \ k,i=1,2,3\end{aligned} \tag{4}$$

where $\nabla_i(\cdot)$ denotes the gradient tensor along the ith-mode. Then the augmented Lagrangian function associated with (4) is given as follows:

$$\begin{aligned}&L\left(\mathcal{X},\mathcal{G}_{ki},\mathcal{N},\mathcal{S},\mathcal{M}_{ki},\mathcal{Z},\mu,\beta\right)\\ &=\lambda_1\|\mathcal{N}\|_F^2+\lambda_2 G_2\left(\mathcal{S}\right)\\ &\quad+\sum_{i=1}^{3}\sum_{k=1}^{3}\Big(\tfrac{1}{3}\alpha_k \text{NP}_k\left(\mathcal{G}_{ki}\right)\\ &\qquad\qquad+\langle\nabla_i\left(\mathcal{X}\right)-\mathcal{G}_{ki},\mathcal{M}_{ki}\rangle+\tfrac{\mu}{2}\|\nabla_i\left(\mathcal{X}\right)-\mathcal{G}_{ki}\|_F^2\Big)\\ &\quad+\langle\mathcal{Y}-\mathcal{X}-\mathcal{N}-\mathcal{S},\mathcal{Z}\rangle+\tfrac{\beta}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{N}-\mathcal{S}\|_F^2\end{aligned} \tag{5}$$

where $\mu,\beta>0$ are given penalty parameters, $\mathcal{M}_{ki}$ and $\mathcal{Z}$ are Lagrange multipliers. The iteration framework of ADMM is given as follows:

$$\begin{cases}\mathcal{X}^{t+1}=\arg\min\limits_{\mathcal{X}} L\left(\mathcal{X},\mathcal{G}_{ki}^{t},\mathcal{N}^{t},\mathcal{S}^{t},\mathcal{M}_{ki}^{t},\mathcal{Z}^{t},\mu^{t},\beta^{t}\right)\\ \mathcal{G}_{ki}^{t+1}=\arg\min\limits_{\mathcal{G}_{ki}} L\left(\mathcal{X}^{t+1},\mathcal{G}_{ki},\mathcal{N}^{t},\mathcal{S}^{t},\mathcal{M}_{ki}^{t},\mathcal{Z}^{t},\mu^{t},\beta^{t}\right)\\ \mathcal{N}^{t+1}=\arg\min\limits_{\mathcal{N}} L\left(\mathcal{X}^{t+1},\mathcal{G}_{ki}^{t+1},\mathcal{N},\mathcal{S}^{t},\mathcal{M}_{ki}^{t},\mathcal{Z}^{t},\mu^{t},\beta^{t}\right)\\ \mathcal{S}^{t+1}=\arg\min\limits_{\mathcal{S}} L\left(\mathcal{X}^{t+1},\mathcal{G}_{ki}^{t+1},\mathcal{N}^{t+1},\mathcal{S},\mathcal{M}_{ki}^{t},\mathcal{Z}^{t},\mu^{t},\beta^{t}\right)\\ \mathcal{M}_{ki}^{t+1}=\mathcal{M}_{ki}^{t}+\mu^{t}\left(\nabla_i\left(\mathcal{X}^{t+1}\right)-\mathcal{G}_{ki}^{t+1}\right)\\ \mathcal{Z}^{t+1}=\mathcal{Z}^{t}+\beta^{t}\left(\mathcal{Y}-\mathcal{X}^{t+1}-\mathcal{N}^{t+1}-\mathcal{S}^{t+1}\right)\\ \mu^{t+1}=\min\left\{\rho\mu^{t},\mu_{\max}\right\},\beta^{t+1}=\min\left\{\rho\beta^{t},\beta_{\max}\right\}\end{cases} \tag{6}$$

where $\rho>1,\beta_{\max}$, and $\mu_{\max}>0$ are given constants.

For the subproblem with respect to $\mathcal{X}$ at the $(t+1)$th iteration, it can be equivalently formulated as follows:

$$\begin{aligned}\mathcal{X}^{t+1}=\arg\min_{\mathcal{X}}&\sum_{i=1}^{3}\sum_{k=1}^{3}\frac{\mu^t}{2}\left\|\nabla_i\left(\mathcal{X}\right)-\mathcal{G}_{ki}^{t}+\frac{\mathcal{M}_{ki}^{t}}{\mu^t}\right\|_F^2\\ &+\frac{\beta^t}{2}\left\|\mathcal{Y}-\mathcal{X}-\mathcal{N}^t-\mathcal{S}^t+\frac{\mathcal{Z}^t}{\beta^t}\right\|_F^2.\end{aligned} \tag{7}$$

The optimal solution of (7) is equivalent to solve the following equation:

$$\begin{aligned}&\left(\beta^t\mathcal{I}+3\mu^t\sum_{i=1}^{3}\nabla_i^T\nabla_i\right)\left(\mathcal{X}\right)\\ &=\beta^t\left(\mathcal{Y}-\mathcal{N}^t-\mathcal{S}^t-\frac{\mathcal{Z}^t}{\beta^t}\right)+\mu^t\sum_{i=1}^{3}\sum_{k=1}^{3}\nabla_i^T\left(\mathcal{G}_{ki}^{t}-\frac{\mathcal{M}_{ki}^{t}}{\mu^t}\right)\end{aligned} \tag{8}$$

where $\mathcal{I}$ denotes the identity operator and ∇_i^T denotes the transpose operator of $\nabla_i(\cdot)$. Let $\mathcal{H}:=\beta^t(\mathcal{Y}-\mathcal{N}^t-\mathcal{S}^t-\mathcal{Z}^t/\beta^t)+\mu^t\sum_{i=1}^{3}\sum_{k=1}^{3}\nabla_i^T(\mathcal{G}_{ki}^t-\mathcal{M}_{ki}^t/\mu^t)$. By the block circulant structure of the matrix corresponding to the operator $\nabla_i^T\nabla_i$, it can be diagonalized by the 3-D FFT matrix [55]. Therefore, we get the following:

$$\mathcal{X}^{t+1}=\mathcal{F}^{-1}\left(\frac{\mathcal{F}\left(\mathcal{H}\right)}{\beta^t\mathbb{I}+3\mu^t\sum_{i=1}^{3}\left|\mathcal{F}\left(\nabla_i\right)\right|^2}\right) \tag{9}$$

where $\mathbb{I}$ represents a tensor with all element being 1, $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the 3-D Fourier transform and its inverse transform, respectively, $|\cdot|^2$ is the element-wise square, and the division is also performed in an element-wise manner.

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For the sub-problem with respect to $\mathcal{G}_{ki}$ at the $(t+1)$th iteration, it can be equivalently formulated as

$$\mathcal{G}_{ki} = \arg\min_{\mathcal{G}_{ki}} \frac{1}{3}\alpha_k \mathrm{NP}_k(\mathcal{G}_{ki}) + \frac{\mu^t}{2}\left\|\mathcal{G}_{ki} - \left(\nabla_i(\mathcal{X}^t) + \frac{\mathcal{M}_{ki}^t}{\mu^t}\right)\right\|_F^2 \tag{10}$$

whose optimal solution is given in the following theorem.

Theorem 1: For any $\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mu > 0$, where $\mathcal{B} = \mathcal{U}_k \diamond_k \mathcal{S}_k \diamond_k \mathcal{V}_k^{T_k}$, the minimizer of the following problem:

$$\min_{\mathcal{A}} \ \mu \mathrm{NP}_k(\mathcal{A}) + \frac{1}{2}\|\mathcal{A} - \mathcal{B}\|_F^2$$

is given by $\mathcal{U}_k \diamond_k \Sigma_k \diamond_k \mathcal{V}_k^{T_k}$, where $\mathrm{bdiag}(\widehat{(\Sigma_k)}_k, k) = \mathrm{Diag}(\mathrm{Prox}_{\mu g_1}(s_1), \ldots, \mathrm{Prox}_{\mu g_1}(s_{\tilde{n}}))$ with s_i being the ith diagonal element of the diagonal matrix $\mathrm{bdiag}(\widehat{(\mathcal{S}_k)}_k, k)$. Here $\tilde{n} := \min\{n_i n_k, n_j n_k\}$ and $i \neq j \neq k, i, j, k = 1, 2, 3$.

The proof of Theorem 1 is similar to that of [40, Theorem 2], which uses the mode-k tensor SVD instead of using mode-3 tensor SVD in [40]. For brevity, we omit the detailed proof here. Let the mode-k tensor SVD of $\nabla_i(\mathcal{X}^t) + \mathcal{M}_{ki}^t/\mu^t$ be $\mathcal{U}_{ki} \diamond_k \mathcal{S}_{ki} \diamond_k \mathcal{V}_{ki}^{T_k}$. Therefore, it follows from Theorem 1 that the optimal solution of (10) is given by

$$\mathcal{G}_{ki}^{t+1} = \mathcal{U}_{ki} \diamond_k \Sigma_{ki} \diamond_k \mathcal{V}_{ki}^{T_k} \tag{11}$$

where

$$\mathrm{bdiag}(\widehat{(\Sigma_{ki})}_k, k) = \mathrm{Diag}\left(\mathrm{Prox}_{\frac{\alpha_k}{3\mu^t} g_1}(s_1), \ldots, \mathrm{Prox}_{\frac{\alpha_k}{3\mu^t} g_1}(s_{\tilde{n}})\right)$$

with s_j being the jth diagonal element of the diagonal matrix $\mathrm{bdiag}(\widehat{(\mathcal{S}_{ki})}_k, k)$.

For the sub-problem with respect to $\mathcal{N}$ at the $(t+1)$th iteration, it can be equivalently formulated as follows:

$$\mathcal{N}^{t+1} = \arg\min_{\mathcal{N}} \lambda_1 \|\mathcal{N}\|_F^2 + \frac{\beta^t}{2}\left\|\mathcal{Y} - \mathcal{X}^{t+1} - \mathcal{N} - \mathcal{S}^t + \frac{\mathcal{Z}^t}{\beta^t}\right\|_F^2 \tag{12}$$

whose closed-form solution is given as follows:

$$\mathcal{N}^{t+1} = \frac{\beta^t\left(\mathcal{Y} - \mathcal{X}^{t+1} - \mathcal{S}^t\right) + \mathcal{Z}^t}{2\lambda_1 + \beta^t}. \tag{13}$$

For the sub-problem with respect to $\mathcal{S}$ at the $(t+1)$th iteration, it can be equivalently formulated as follows:

$$\mathcal{S}^{t+1} = \arg\min_{\mathcal{S}} \lambda_2 G_2(\mathcal{S}) + \frac{\beta^t}{2}\left\|\mathcal{S} - \left(\mathcal{Y} - \mathcal{X}^{t+1} - \mathcal{N}^t + \frac{\mathcal{Z}^t}{\beta^t}\right)\right\|_F^2. \tag{14}$$

The solution of problem (14) is the proximal mapping of $\lambda_2/\beta^t G_2$, which can be computed in an element-wise manner [40] and given as follows:

$$\left(\mathcal{S}^{t+1}\right)_{ljm} = \mathrm{Prox}_{\frac{\lambda_2}{\beta^t} g_2}\left(\left(\mathcal{Y} - \mathcal{X}^{t+1} - \mathcal{N}^t + \frac{\mathcal{Z}^t}{\beta}\right)_{ljm}\right). \tag{15}$$

Now we state the ADMM for solving model (4) in Algorithm 1. The convergence result and computational complexity of Algorithm 1 are given in Supplementary Material.

Algorithm 1 ADMM for Solving Model (4)

Input: Noisy HSI $\mathcal{Y}$; parameters $\alpha_k, \lambda_1, \lambda_2, \mu^0, \beta^0$; $\rho > 1, \beta_{\max}, \mu_{\max}$; initial tensors $\mathcal{G}_{ki}^0, \mathcal{S}^0, \mathcal{N}^0, \mathcal{M}_{ki}^0, \mathcal{Z}^0$.
Output: Restored HSI $\mathcal{X}^{t+1}$.

```
for t = 0, 1, ... do
    Update X^{t+1} via (9).
    Update G_{ki}^{t+1} (k, j = 1, 2, 3) via (11).
    Update N^{t+1} via (13).
    Update S^{t+1} via (15).
    Update M_{ki}^{t+1} and Z^{t+1} via (6).
    β^{t+1} = min{ρβ^t, β_max}; μ^{t+1} = min{ρμ^t, μ_max}.
    if ||X^{t+1} − X^t||_F / ||X^t||_F ≤ 10^{-3} then
        break
    end if
end for
```

During the implementation of Algorithm 1, one needs to compute the proximal mappings of g_1 and g_2. In particular, when g_1 is the MCP, the proximal mapping of μg_1 is given [40] as follows:

$$\mathrm{Prox}_{\mu g_1}(y) = \begin{cases} 0, & \text{if } |y| \leqslant c\mu \\ \dfrac{\mathrm{sign}(y)(|y| - c\mu)}{1 - \mu/\eta}, & \text{if } c\mu < |y| \leqslant c\eta \\ y, & \text{if } |y| > c\eta \end{cases}$$

where $0 < \mu < \eta$.

V. Experimental Results

In this section, some numerical experiments on both simulated and real-world HSI datasets are conducted. To demonstrate the effectiveness of 3DTCTV and 3DNP methods, we compare with the following methods for removing mixed noises of HSIs: TV-regularized low-rank matrix factorization (LRTV) [20], nonlocal meets global method (NGmeet) [18], TV-regularized low-rank tensor decomposition (LRTDTV) [55], three-directional TNN with logarithmic penalty (3DLogTNN) [77], factor group sparsity-regularized nonconvex low-rank approximation (FGSLR) [8], representative coefficient total variation (RCTV) [37], reconcile sparse and low-tensor-ring-rank priors in the learned transformed domain (TRSTR) [71], and fast graph Laplacian regularizer method (FGLR) [48]. All parameters in the compared methods are set according to the suggestions of the corresponding literature. Moreover, the testing HSIs are normalized to $[0, 1]$ band-by-band. All experiments are performed on MATLAB R2017b using a desktop of 16 GB RAM with an Intel Core i7-8700K CPU at 3.70 GHz.

For the unitary matrices in 3DTCTV and 3DNP, we will give the choices and discuss different transformations for HSI denoising in detail in Section V-C.2. In the 3DNP model, we use the MCP function for both g_1 and g_2. In Section V-C.3, we will analyze the denoising performance of different nonconvex functions of g_1 and g_2 by numerical experiments. For the initial tensors $\mathcal{G}_{ki}^0, \mathcal{S}^0, \mathcal{N}^0, \mathcal{M}_{ki}^0$, and $\mathcal{Z}^0$ of ADMM in Algorithm 1, we generate $\mathcal{S}^0, \mathcal{M}_{ki}^0$, and $\mathcal{N}^0$ as zero tensors, and $\mathcal{G}_{ki}^0$ and $\mathcal{N}^0$ as random tensors distributed on [0,1] uniformly.

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TABLE II
MPSNR, MSSIM, AND SAM OF DIFFERENT METHODS FOR THE PAVIA CITY CENTER DATASET WITH DIFFERENT NOISE CASES

Case	Metric	Noisy	LRTV	NGmeet	LRTDTV	3DLogTNN	FGLSR	RCTV	TRSTR	FGLR	3DTCTV	3DNP
Case 1	MPSNR	11.20	23.59	26.66	32.30	33.91	34.04	33.68	32.89	31.24	33.47	**34.62**
	MSSIM	0.0924	0.7198	0.7391	0.8648	**0.9431**	0.9061	0.8953	0.8806	0.8728	0.9073	0.9194
	SAM	43.24	7.6376	5.1718	6.0616	**3.2263**	3.3514	4.0127	4.0413	5.4129	3.6997	3.3454
Case 2	MPSNR	10.58	23.01	26.08	30.23	30.73	31.68	31.07	30.84	29.18	31.47	**32.70**
	MSSIM	0.0794	0.6944	0.6993	0.7940	0.8665	0.8599	0.8322	0.8299	0.8062	0.8659	**0.8904**
	SAM	45.4416	9.8511	5.6177	8.3286	4.0153	4.0306	6.2932	4.8922	6.6896	4.6804	**4.0013**
Case 3	MPSNR	9.83	21.78	24.65	29.92	30.12	31.18	30.42	30.23	28.76	30.87	**32.23**
	MSSIM	0.0632	0.6453	0.6764	0.7867	0.8534	0.8472	0.8230	0.8142	0.7946	0.8536	**0.8819**
	SAM	46.4772	9.5712	6.7210	8.9064	4.2681	4.1587	6.6609	5.0583	6.8649	4.9816	**4.1052**
Case 4	MPSNR	9.76	21.31	24.64	29.78	30.02	31.03	30.46	30.09	28.63	30.66	**32.08**
	MSSIM	0.0618	0.6424	0.6668	0.7795	0.8496	0.8442	0.8174	0.8107	0.7881	0.8483	**0.8788**
	SAM	46.5968	11.0533	6.4330	8.5845	4.2973	**4.2498**	6.4384	5.1903	7.0180	5.1207	4.2726
Case 5	MPSNR	9.94	21.77	24.94	29.86	29.74	30.96	30.78	30.35	28.80	30.89	**32.45**
	MSSIM	0.0653	0.6592	0.6750	0.7817	0.8384	0.8384	0.8236	0.8163	0.7970	0.8549	**0.8857**
	SAM	46.4773	11.1701	6.2575	9.5416	5.3646	6.2134	6.6559	5.2175	6.9869	5.0017	**3.9150**
Case 6	MPSNR	9.75	21.46	24.59	29.43	29.03	30.80	30.01	29.95	28.53	30.39	**32.06**
	MSSIM	0.0616	0.6438	0.6692	0.7705	0.8172	0.8385	0.8111	0.8074	0.7857	0.8450	**0.8787**
	SAM	46.8915	11.2186	7.0532	9.2535	6.2207	4.5254	7.1894	5.4056	7.0992	5.3026	**4.1630**
	Time		46.79	111.35	120.21	413.64	140.58	**3.23**	29.07	5.36	515.26	548.31

A. Simulated Experiments

In this section, we conduct the experiments on the Pavia City Center dataset[1] ($200 \times 200 \times 80$) and Washington DC Mall dataset[2] ($256 \times 256 \times 191$), where the clean HSIs of the two datasets can be obtained. The mean peak signal-to-noise ratio (MPSNR), mean structural similarity index (MSSIM) [56], and spectral angle mapping (SAM) are used to measure the restoration quality of different methods for the simulated experiments. Moreover, higher MPSNR, MSSIM, and lower SAM values correspond to better denoising performances.

For the simulated experiments, we consider the following six cases of different mixture noises.

1) *Case 1. Independent identically distributed (i.i.d.) Gaussian Noise + i.i.d. Impulse Noise.* All bands are corrupted by Gaussian noise with zero mean and standard deviation $\sigma = 0.1$. Additionally, impulse noise is added to all bands with noise level $v = 0.2$ for each band.
2) *Case 2. Non-i.i.d. Gaussian Noise + i.i.d. Impulse Noise.* The impulse noise is the same as in *Case 1*. The zero mean Gaussian noise is also added with the standard deviation σ randomly selecting from $[0.1, 0.2]$ for each band.
3) *Case 3. Non-i.i.d. Gaussian Noise + Non-i.i.d. Impulse Noise.* The Gaussian noise is the same as in *Case 2*. The impulse noise is added to all bands with the percentages of polluted pixels randomly selected from $[0.2, 0.3]$ for each band.
4) *Case 4. Non-i.i.d. Gaussian Noise + Non-i.i.d. Impulse Noise + Stripe.* The Gaussian noise and impulse noise are added as the same in *Case 3*. Additionally, 15 bands are randomly selected, and each band is randomly corrupted from 5 to 10 stripes.
5) *Case 5. Non-i.i.d. Gaussian Noise + Non-i.i.d. Impulse Noise + Deadline.* The Gaussian noise and impulse noise are added as the same in *Case 3*. Additionally, 15 bands are randomly selected and 5–10 deadlines are randomly added for these selected bands, where the width of the deadline is randomly chosen from $\{1, 2, 3\}$.
6) *Case 6. Non-i.i.d. Gaussian Noise + Non-i.i.d. Impulse Noise + Stripe + Deadline.* The Gaussian noise and impulse noise are added as the same in *Case 3*. The stripe and deadline are added as the same in *Case 4* and *Case 5*, respectively.

In Tables II and III, we report the MPSNR, MSSIM, and SAM values of different methods for the Pavia City Center and Washington DC Mall datasets under various noise conditions, respectively. The best results are highlighted in bold and the second best results are underlined. It can be seen that the MPSNR and MSSIM values of 3DNP are higher than other methods, and the SAM values of 3DNP are smaller than those of other methods for most cases, which implies that the 3DNP outperforms other methods for HSI denoising in the testing cases. Moreover, in most cases, the denoising performances of FGLSR and 3DTCTV are better than those of LRTV, NGmeet, LRTDTV, 3DLogTNN, TRSTR, and FGLR for the Pavia City Center dataset, and the denoising performances of RCTV and 3DTCTV are better than those of LRTV, NGmeet, LRTDTV, 3DLogTNN, FGLSR, TRSTR, and FGLR for the Washington DC Mall dataset in terms of MPSNR, MSSIM, and SAM. Although the average time (in seconds) of 3DTCTV and 3DNP is higher than that of other methods, the recovery performance of our methods is better for HSI denoising.

In Fig. 2, we show the visual comparisons of different methods for bands (11, 54, 22) of the Pavia City Center dataset and bands (134, 16, 82) of the Washington DC Mall dataset in Case 6, respectively. The zoomed regions of these images are also displayed. From the top row of Fig. 2, we can observe that all methods can remove the mixture noises to some extent. However, the FGLSR cannot remove the stripe noise. Moreover, the LRTV, NGmeet, LRTDTV, 3DLogTNN, TRSTR, and FGLR cannot remove the noise efficiently, where the zoomed regions of the restored images contain noises. And the 3DNP can remove the mixed noises

[1]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes
[2]http://lesun.weebly.com/hyperspectral-data-set.html

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TABLE III
MPSNR, MSSIM, AND SAM OF DIFFERENT METHODS FOR THE WASHINGTON DC MALL DATASET WITH DIFFERENT NOISE CASES

Case	Metric	Noisy	LRTV	NGmeet	LRTDTV	3DLogTNN	FGLSR	RCTV	TRSTR	FGLR	3DTCTV	3DNP
Case 1	MPSNR	10.70	20.43	24.656	33.64	34.00	34.96	35.65	32.93	27.44	35.04	**36.43**
	MSSIM	0.0779	0.1366	0.7820	0.8263	0.8814	0.8893	0.8964	0.8395	0.6887	0.9131	**0.9194**
	SAM	43.2370	15.6116	13.5423	5.8489	5.0548	5.2142	4.7424	6.1232	14.7022	4.6961	**4.1968**
Case 2	MPSNR	10.12	20.42	24.55	32.09	31.82	32.60	32.88	31.41	27.14	32.77	**34.63**
	MSSIM	0.0666	0.1356	0.7755	0.7777	0.8357	0.8436	0.8415	0.8035	0.6759	0.8439	**0.8904**
	SAM	52.6590	15.6771	13.5514	7.3422	6.4770	6.5888	6.3384	7.0372	14.9115	6.0645	**5.0768**
Case 3	MPSNR	9.38	19.19	22.77	31.57	30.98	31.99	32.09	30.74	26.81	32.07	**34.18**
	MSSIM	0.0544	0.1147	0.7229	0.7617	0.8194	0.8283	0.8284	0.7903	0.6620	0.8294	**0.8819**
	SAM	53.4839	17.4007	15.6380	7.7772	7.2873	7.0684	7.0621	7.6056	15.2371	6.6965	**5.3510**
Case 4	MPSNR	9.29	19.09	22.61	31.43	30.76	31.64	31.77	30.51	26.70	31.79	**33.97**
	MSSIM	0.0530	0.1139	0.7217	0.7565	0.8144	0.8081	0.8183	0.7845	0.6584	0.8238	**0.8788**
	SAM	53.6276	17.5008	15.7821	7.9741	7.5106	8.0912	7.3664	7.7689	15.3784	6.9411	**5.5050**
Case 5	MPSNR	9.33	19.15	22.67	31.48	30.77	31.94	32.00	30.68	26.89	31.92	**34.04**
	MSSIM	0.0535	0.1142	0.7209	0.7570	0.8099	0.8228	0.8235	0.7885	0.6641	0.8258	**0.8857**
	SAM	53.7001	17.6486	15.9647	8.1500	7.6619	7.2266	7.1378	7.7084	15.1508	6.8030	**5.4289**
Case 6	MPSNR	9.26	19.07	22.59	31.32	30.57	31.67	31.72	30.49	26.74	31.66	**33.87**
	MSSIM	0.0519	0.1117	0.7170	0.7523	0.8056	0.8134	0.8144	0.7830	0.6573	0.8203	**0.8633**
	SAM	53.7594	17.7151	15.9858	7.9924	7.6284	7.4063	7.2060	7.7844	15.2132	6.9156	**5.5112**
	Time		301.50	334.45	372.72	1505.53	1192.71	**11.27**	76.43	21.76	2029.77	2151.09

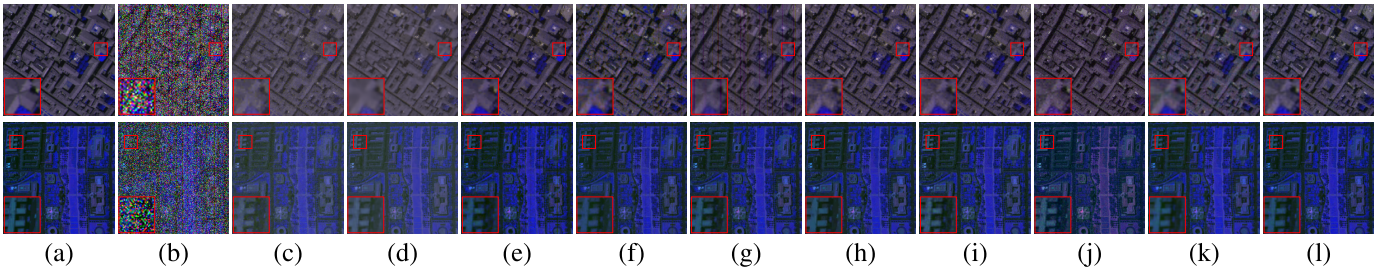


Fig. 2. (Top row) Restoration results for bands (11, 54, 22) of the Pavia City Center dataset (Case 6). (Bottom row) Restoration results for bands (134, 16, 82) of the Washington DC Mall dataset (Case 6). (a) Original. (b) Noisy. (c) LRTV. (d) NGmeet. (e) LRTDTV. (f) 3DLogTNN. (g) FGLSR. (h) RCTV. (i) TRSTR. (j) FGLR. (k) 3DTCTV. (l) 3DNP.

and preserve the local details better than other methods. From the bottom row of Fig. 2, we can see that LRTDTV and 3DLogTNN cannot remove the stripes efficiently. Furthermore, the images recovered by LRTV, NGmeet, FGLSR, TRSTR, and FGLR contain some Gaussian noises, and the corners of HSIs recovered by RCTV are not clear, while 3DTCTV and 3DNP can remove the mixture noise and preserve the local details of HSIs well.

In Fig. 3, we show that the PSNR and SSIM values of different methods versus index of band for the Pavia City Center and Washington DC Mall datasets in Case 3 and Case 6, respectively. It can be seen that the 3DNP displays higher and more stable PSNR and SSIM values compared with LRTV, NGmeet, LRTDTV, 3DLogTNN, FGLSR, RCTV, TRSTR, FGLR, and 3DTCTV in most bands for different noise cases and HSIs, which implies that the 3DNP can obtain better denoising performance than other comparison methods.

In order to analyze the spectral recovery capability of each method, we compared the spectral curves of different methods. In Fig. 4, we show the original spectral signature (SS) curves (blue curve) and denoised SS curves (red curve) of different methods at spatial pixel (120, 120) for the Pavia City Center dataset and (145,120) for the Washington DC Mall dataset, respectively. It is evident that the spectral curves obtained by 3DNP are most consistent with the original spectral curves, while the other methods are more or less impacted by noise.

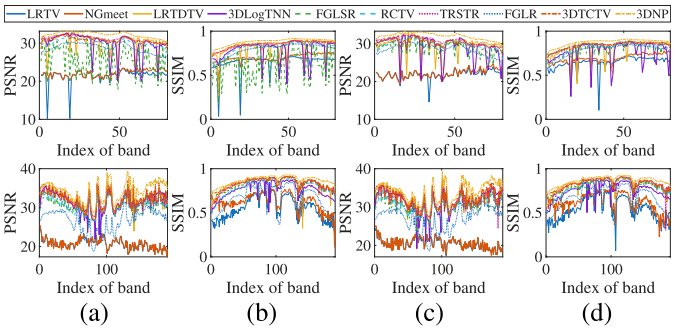


Fig. 3. (Top row) PSNR and SSIM values versus index of band for the Pavia City Center dataset (Case 3 and Case 6). (Bottom row) PSNR and SSIM values versus index of band for the Washington DC Mall dataset (Case 3 and Case 6). (a) Case 3: PSNR versus index of band. (b) Case 3: SSIM versus index of band. (c) Case 6: PSNR versus index of band. (d) Case 6: SSIM versus index of band.

These results regarding the spectral curves of different datasets further demonstrate the superiority and robustness of the 3DNP method.

B. Real Experiments

In this section, we test two real-world HSIs, that is, Indian Pines dataset[3] (145 × 145 × 220) and WHU-Hi-HongHu

[3]https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

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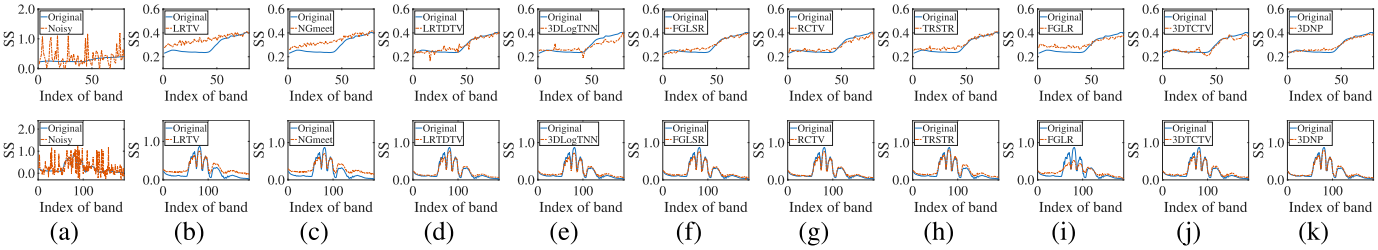


Fig. 4. (Top row) Spatial-spectrum results for the Pavia City Center dataset (spatial pixel (120, 120) in Case 6). (Bottom row) Spatial-spectrum results for Washington DC Mall (spatial pixel (145, 120) in Case 6). (a) Noisy. (b) LRTV. (c) NGmeet. (d) LRTDTV. (e) 3DLogTNN. (f) FGLSR. (g) RCTV. (h) TRSTR. (i) FGLR. (j) 3DTCTV. (k) 3DNP.

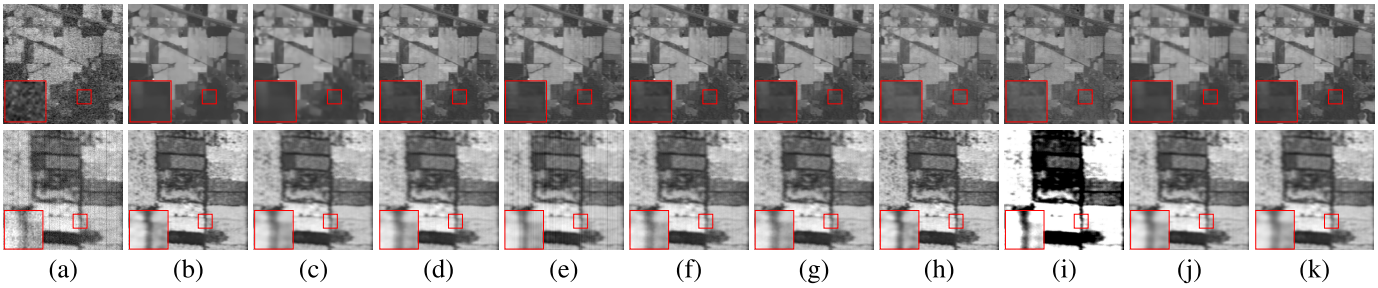


Fig. 5. (Top row) Restoration results of different methods for the Indian Pines dataset in band 163. (Bottom row) Restoration results of different methods for the WHU-Hi-HongHu dataset in band 245. (a) Noisy. (b) LRTV. (c) NGmeet. (d) LRTDTV. (e) 3DLogTNN. (f) FGLSR. (g) RCTV. (h) TRSTR. (i) FGLR. (j) 3DTCTV. (k) 3DNP.

dataset.[4] Since the spatial dimension of the WHU-Hi-HongHu dataset ($940 \times 475 \times 270$) is too large, we choose a subimage with size $256 \times 256 \times 270$ to demonstrate the effectiveness of the proposed methods. The Indian Pines dataset is severely corrupted by dense Gaussian noise, impulse noise, stripes, and deadlines. The WHU-Hi-HongHu dataset is seriously corrupted by Gaussian noise, stripes, impulse noise, and deadlines. Since there are no clean images for comparisons, we can only judge the denoising quality based on experiences. For denoised images, if an approach can better preserve color, texture, and local smoothness of the image, and remove more noise, it can be considered to achieve better denoising results. In the top row of Fig. 5, we show the visual denoising images of band 163 of different methods for the Indian Pines dataset. It can be observed that the images recovered by LRTV are overly smoothing. And the NGmeet, LRTDTV, 3DLogTNN, FGLSR, RCTV, TRSTR, and FGLR cannot completely reduce the hybrid noise or cannot finitely preserve the structures of images. In contrast, 3DTCTV and 3DNP can preserve the structures of the recovered images better and remove the hybrid noise efficiently, which demonstrate that the proposed methods can obtain better visual performance.

In the bottom row of Fig. 5, we show the visual quality of the recovered images of band 245 of different methods for the WHU-Hi-HongHu dataset. From this figure, we can observe that the 3DLogTNN cannot remove the stripe noise, and the FGLR cannot preserve the structures of images. Moreover, LRTV, NGmeet, LRTDTV, FGLSR, RCTV, and TRSTR cannot finitely remove the noise. By comparison, 3DTCTV and 3DNP can remove the hybrid noise of WHU-Hi-HongHu and preserve the local details better. From the denoised images, it is evident that the 3DNP has more discriminative features in terms of noise removal and structural preservation.

Regarding the spectral curves, if a method can denoise the spectral curve while effectively preserving the trend of the original spectral curve of noisy HSI (such as local smoothness) and still maintaining the characteristics of a specific point, then this method can be considered effective for denoising. In Fig. 6, the SS curves at spatial pixels (80, 80) and (83, 68) are shown for the Indian Pines and WHU-Hi-HongHu datasets, respectively. We can see from the two figures that our methods can preserve the structure of spectral curves and remove the noise better than other comparison methods.

C. Discussion

1) Convergence Curves: In this section, we present the convergence curve to show the convergence of Algorithm 1. The MPSNR, MSSIM, SAM versus number of iterations of 3DNP are displayed in Fig. 7 for the Pavia City Center and Washington DC Mall datasets in Case 3. It can be seen that the MPSNR and MSSIM values increase and the SAM values decrease as the number of iterations increases, and then these metric values tend toward stability when the number of iterations is larger than 30, which demonstrate the convergence behavior of Algorithm 1 as shown in Theorem 1 in Supplementary Material.

2) Performance of Different Transformations: In this section, we test the performance of 3DTCTV and 3DNP with different transformations for the Pavia City Center dataset. Here we use two fixed transformations (i.e., fast Fourier transform (FFT) and DCT) and a data-dependent transformation, where the three methods are called 3DTCTV (FFT), 3DTCTV (DCT), 3DTCTV (Data) and 3DNP (FFT), 3DNP (DCT), 3DNP (Data), respectively. For the data-dependent

[4]https://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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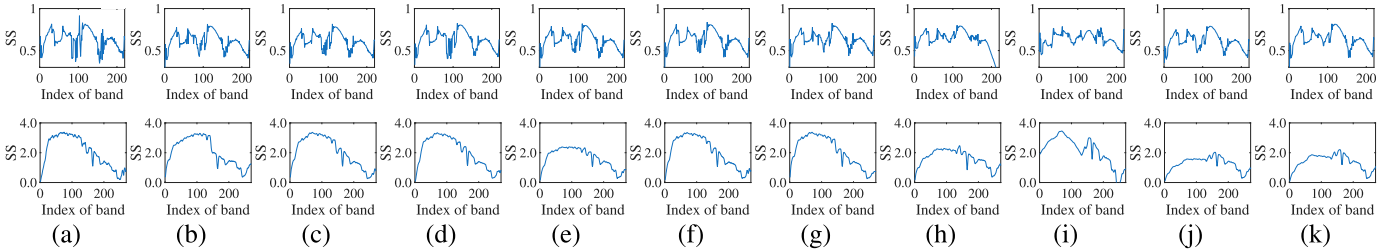


Fig. 6. (Top row) Original spectrum and the denoising results of different methods at spatial pixel (80, 80) for the Indian Pines dataset. (Bottom row) Original spectrum and the denoising results of different methods at spatial pixel (83, 68) for the WHU-Hi-HongHu dataset. (a) Noisy. (b) LRTV. (c) NGmeet. (d) LRTDTV. (e) 3DLogTNN. (f) FGLSR. (g) RCTV. (h) TRSTR. (i) FGLR. (j) 3DTCTV. (k) 3DNP.

TABLE IV
MPSNR, MSSIM, AND SAM VALUES OF 3DTCTV AND 3DNP WITH DIFFERENT TRANSFORMATIONS FOR THE PAVIA CITY CENTER DATASET

		3DTCTV			3DNP		
		FFT	DCT	Data	FFT	DCT	Data
Case 1	MPSNR	32.93	32.99	33.47	33.67	33.72	34.62
	MSSIM	0.9333	0.9324	0.9073	0.9401	0.9402	0.9194
	SAM	3.7553	3.9472	3.6997	3.6290	3.3815	3.3454
Case 2	MPSNR	31.00	31.07	31.47	31.42	31.93	32.70
	MSSIM	0.9013	0.9004	0.8659	0.9052	0.9135	0.8904
	SAM	4.5377	4.8738	4.6804	4.9723	4.0846	4.0013

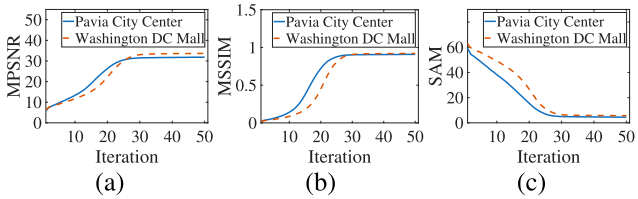


Fig. 7. MPSNR, MSSIM, and SAM values of 3DNP versus number of iterations for the Pavia City Center and Washington DC Mall datasets in Case 3. (a) MPSNR versus number of iterations. (b) MSSIM versus number of iterations. (c) SAM versus number of iterations.

transformation, an initial estimator $\widetilde{\mathcal{X}}$ based 3DTCTV with DCT or 3DNP with DCT is generated. Then we take the gradient of $\widetilde{\mathcal{X}}$ along the i-mode, which is denoted by $\mathcal{D}_i, i = 1, 2, 3$. Then we unfold $\mathcal{D}_i$ along the j-dimension into a matrix $\mathbf{D}_{ij}, j = 1, 2, 3$. By taking the SVD of $\mathbf{D}_{ij}$ as $\mathbf{D}_{ij} = \mathbf{U}_{ij}\Sigma_{ij}\mathbf{V}_{ij}^T$. Then $\mathbf{U}_{ij}^T$ is a desirable unitary for mode-$j$tensor SVD of the ith gradient tensor, which can generate a low-rank tensor. The interested reader can refer to [45] and [46] for more details regarding the choose of unitary matrix in transformed tensor SVD.

In Table IV, we show the MPSNR, MSSIM, SAM values of 3DTCTV and 3DNP with different transformations for the Pavia City Center dataset. It can be observed that the performance of DCT and FFT is almost the same for 3DTCTV, which is slightly worse than that of data-dependent transformation for both Case 1 and Case 2. Moreover, the 3DNP with DCT and data-dependent transformations performs better than that with FFT for the testing two cases, while the 3DNP with DCT performs almost the same as that with the data-dependent transformation. In the experiments, we choose the data-dependent transformation for both 3DTCTV and 3DNP.

3) Performance of Different Nonconvex Functions: In this section, we test the denoising performance of different nonconvex functions for 3DNP, where g_1 and g_2 are chosen as the same for simplicity in the experiments. We consider the ℓ_p norm, SCAD, logarithmic function, and MCP in Table I for g_1, g_2, and also compare with 3DTCTV to demonstrate the effectiveness of the nonconvex method, where the ℓ_1 norm is used for the low-rank and sparse components in 3DTCTV. The corresponding methods are called 3DNP (ℓ_p), 3DNP (SCAD), 3DNP (Log), and 3DNP (MCP).

In Table V, we show the MPSNR, MSSIM, SAM of 3DTCTV (FFT), 3DTCTV (DCT), 3DTCTV (Data), 3DNP (ℓ_p), 3DNP (SCAD), 3DNP (Log), and 3DNP (MCP) for the Pavia City Center dataset in Case 6. It can be seen from this table that the 3DNP (ℓ_p), 3DNP (SCAD), 3DNP (Log), and 3DNP (MCP) perform better than 3DTCTV based on FFT, DCT, and data-dependent transformation, which demonstrates the usefulness of the nonconvex functions. And the 3DTCTV (Data) outperforms 3DTCTV (FFT) and 3DTCTV (DCT) slightly in terms of MPSNR and MSSIM values, which shows the superior performance of data-dependent transformation. Furthermore, the performance of 3DNP (ℓ_p), 3DNP (SCAD), and 3DNP (Log) is almost the same for the testing case, while the 3DNP (MCP) is slightly better than other nonconvex methods. For brevity, we choose the MCP function in 3DNP for both g_1 and g_2 in all experiments.

4) Parameters Analysis: In this section, we analyze the parameters ($\alpha_k, k = 1, 2, 3, \lambda_1, \lambda_2$) of the 3DTCTV, 3DNP, and the parameter ρ in Algorithm 1. For the parameters c, η in MCP of 3DNP, they are not too sensitive to the results, we simply choose $c = 0.6$ and $\eta = 2.8$ for all cases. In Algorithm 1, we set $\mu_{\max} = \beta_{\max} = 1000$ in all experiments for both 3DTCTV and 3DNP. For μ^0 in Algorithm 1 for solving 3DTCTV, we choose it as 0.05 and 0.01 for the Pavia City Center and Washington DC Mall datasets, respectively. For β^0 in Algorithm 1 for solving 3DTCTV, we choose it as 0.05 and 0.001 for the Pavia City Center and Washington DC Mall datasets, respectively. In Algorithm 1 for solving 3DNP, μ^0 and β^0 are set as 0.5 and 0.01 for the Pavia City Center and Washington DC Mall datasets, respectively.

Since α_1, α_2 are the two weights of TNN for the spatial dimensions of HSIs, we simply set them the same. Specifically, we set $(\alpha_1, \alpha_2, \alpha_3) = (a, a, b)/(2a+b)$ and analyze the influence of a, b. In Fig. 8, we show the influence of parameters $\lambda_1, \lambda_2, a, b, \rho$ of 3DTCTV and 3DNP for the Pavia City Center dataset under Case 6. For 3DTCTV, λ_1 varies from 0.1 to 1.5 with step size 0.1 and λ_2 varies from 1 to 11 with step size 1. For 3DNP, λ_1 varies from 0.005 to 0.95 with step size 0.01,

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TABLE V

COMPARISONS OF 3DNP WITH DIFFERENT NONCONVEX FUNCTIONS FOR THE PAVIA CITY CENTER DATASET IN CASE 6

	Noisy	3DTCTV (FFT)	3DTCTV (DCT)	3DTCTV (Data)	3DNP (ℓ_p)	3DNP (SCAD)	3DNP (Log)	3DNP (MCP)
MPSNR	9.75	30.01	30.11	30.39	31.55	31.47	31.26	32.06
MSSIM	0.0703	0.8421	0.8440	0.8450	0.9047	0.9063	0.9029	0.8787
SAM	46.8915	5.5253	5.3625	5.3026	4.1814	4.3854	4.5357	4.1630

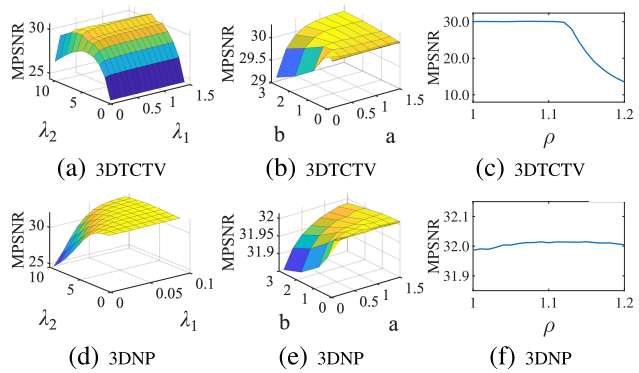


Fig. 8. MPSNR versus different parameters of 3DTCTV and 3DNP for the Pavia City Center dataset under Case 6. (a) and (d) MPSNR versus (λ_1, λ_2). (b) and (e) MPSNR versus (a, b). (c) and (f) MPSNR versus ρ.

TABLE VI

MPSNR, MSSIM, AND SAM VALUES OF DIFFERENT METHODS FOR THE PAVIA CITY CENTER AND WASHINGTON DC MALL (WDC) DATASETS IN CASE 3

		GTNN	G3DTNN	3GTNN	3DTNN	3DTCTV
Pavia	MPSNR	24.78	25.00	28.70	29.16	30.87
	MSSIM	0.6040	0.6129	0.8411	0.8722	0.8536
	SAM	9.9176	9.4651	5.2477	4.7132	4.9816
WDC	MPSNR	25.22	25.64	29.72	29.15	32.07
	MSSIM	0.5573	0.5801	0.8467	0.8493	0.8294
	SAM	13.5104	13.0697	8.9872	8.7761	6.6965
		GTNNNC	G3DTNNNC	3GTNNNC	3DTNNNC	3DNP
Pavia	MPSNR	25.422	26.01	28.95	30.67	32.23
	MSSIM	0.6480	0.6868	0.8430	0.8975	0.8819
	SAM	8.6443	8.6722	5.2509	4.6993	4.1052
WDC	MPSNR	25.98	27.07	30.80	32.32	34.18
	MSSIM	0.6131	0.6616	0.8641	0.9141	0.8819
	SAM	12.3654	10.9809	8.1747	6.0061	5.3510

and λ_2 varies from 1 to 10 with step size 1. From Fig. 8(a) and (d), we can see that λ_1 is stable for the results while λ_2 can be sensitive for both 3DTCTV and 3DNP. In the experiments, for 3DTCTV, we choose $\lambda_1 = 1$ for all experiments, $\lambda_2 = 8$ for the Pavia City Center dataset and $\lambda_2 = 9$ for the Washington DC Mall dataset. For 3DNP, we choose λ_1, λ_2 as 0.075 and 5 for all experiments.

For 3DTCTV and 3DNP, a is chosen from 0.1 to 1.5 with step size 0.2 and b is chosen from the set $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 2, 3\}$. We can see from Fig. 8(b) and (e) that a and b are stable to the results. For simplicity, we set $a = 1$ and $b = 0.001$ in all experiments for both 3DTCTV and 3DNP. In Fig. 8(c) and (f), ρ is chosen from 1 to 1.2 with step size 0.01 for both 3DTCTV and 3DNP. We can see that ρ is not sensitive to the results of for both methods. We set $\rho = 1.1$ for 3DTCTV. For 3DNP, we choose $\rho = 1.1$ for the Pavia City Center dataset and $\rho = 1.2$ for the Washington DC Mall dataset, respectively.

5) Ablation Analysis: In this section, we give the ablation analysis of 3DTCTV and 3DNP. Notice that the 3DTCTV model utilizes three-directional gradient tensors and employs three-directional TNN on each gradient tensor. First, we consider the three-directional TNN for the underlying tensor (called 3DTNN), that is, the 3DTCTV is replaced by 3DTNN in model (2). Moreover, in order to show the effectiveness of 3DTCTV with three-directional gradient tensors and three-directional TNN on each gradient tensor, we compare it with the model using TNN on one gradient tensor (called GTNN). Furthermore, we also compare our model with the three-directional TNN on one gradient tensor (called G3DTNN). Besides, we compare with the model using three-directional gradient with TNN on each gradient tensor (called 3GTNN). The corresponding nonconvex methods are called GTNNNC, G3DTNNNC, 3GTNNNC, and 3DTNNNC, where the MCP function is employed instead of ℓ_1 norm for the low-rank and sparse components in these models.

In Table VI, we report the MPSNR, MSSIM, and SAM values of different methods for the Pavia City Center and Washington DC datasets in Case 3. It can be seen from this table that the MPSNR, MSSIM, and SAM values of G3DTNN and G3DTNNNC are higher than those of GTNN and GTNNNC for different datasets, which shows that the three-directional TNN is more effective than the one-directional TNN on one gradient tensor for both the convex and nonconvex models. Moreover, the 3GTNN and 3GTNNNC outperform GTNN and GTNNNC for these testing datasets, respectively, which demonstrates that the performance using three-directional gradients is better than that using one-directional gradient. Compared with G3DTNN and G3DTNNNC, the 3DTNN and 3DTNNNC perform better in terms of MPSNR, MSSIM, and SAM values. This phenomenon shows that the performance by using three-directional TNN on the original tensor is better than that by using three-directional TNN on one gradient tensor. The performance of 3DTCTV and 3DNP is better than G3DTNN, 3GTNN, and their corresponding nonconvex models, respectively, which demonstrates that the model using three-directional gradients with three-directional TNN on each gradient tensor is better than those using one-directional gradient with three-directional TNN or three-directional gradient tensors with one-directional TNN. Furthermore, the MPSNR and MSSIM values obtained by GTNNNC, G3DTNNNC, 3GTNNNC, 3DTNNNC, and 3DNP are higher and the SAM values of these methods are lower than those obtained by GTNN, G3DTNN, 3GTNN, 3DTNN, and 3DTCTV, which shows that the nonconvex methods are more effective than the corresponding convex methods.

Next we show the usefulness of nonconvex functions on both the low-rank component and sparse component simultaneously. For the model using the nonconvex function

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TABLE VII
MPSNR, MSSIM, AND SAM VALUES OF DIFFERENT METHODS FOR THE PAVIA CITY CENTER DATASET IN CASES 4 AND 5, RESPECTIVELY

		3DTCTV	3DNP-S	3DNP-LR	3DNP
Case 4	MPSNR	30.66	30.99	31.30	32.08
	MSSIM	0.8483	0.8983	0.9044	0.8788
	SAM	5.1207	4.3616	4.4471	4.2726
Case 5	MPSNR	30.89	31.02	31.29	32.45
	MSSIM	0.8549	0.8984	0.9046	0.8857
	SAM	5.0017	4.2825	4.4228	3.9150

only on the low-rank component, that is, the model in (3) uses $\|\mathcal{S}\|_1$ instead of $G_2(\mathcal{S})$, we denote it by 3DNP-LR. Moreover, for the model using the nonconvex function only on the sparse component, that is, the model in (3) uses $\|\mathcal{X}\|_{3\text{DTCTV}}$ instead of $G_1(\mathcal{X})$, we denote it by 3DNP-S. In Table VII, we show the MPSNR, MSSIM, and SAM values of 3DTCTV, 3DNP-S, 3DNP-LR, and 3DNP for the Pavia City Center dataset in Cases 4 and 5, respectively. It can be seen that 3DNP-S and 3DNP-LR outperform 3DTCTV for these testing cases, which shows the usefulness of the nonconvex function on the low-rank component or the sparse component. Moreover, the performance of 3DNP is better than that of 3DNP-S and 3DNP-LR in terms of MPSNR, MSSIM, and SAM, which implies that the nonconvex regularization on both the low-rank component and sparse component simultaneously is better than that on only the low-rank component or sparse component.

VI. CONCLUSION

In this article, we have proposed the 3DTCTV method and its nonconvex relaxation for removing the mixture noises of HSIs. The 3DTCTV utilized the 3-D TTNN onto the gradient tensors of HSIs along different modes, which can explore the low-rankness and local smoothness simultaneously. Then the 3DNP model was proposed to enhance the low-rankness and local smoothness of the clean HSI further, where a family of nonconvex functions were employed onto the singular values of the gradient tensors in the transformed domain along different modes. Moreover, a nonconvex relaxation based on a family of nonconvex functions was employed to approximate the ℓ_0 norm for the sparse noise. An ADMM-based algorithm was developed to solve the resulting models, and the convergence for the nonconvex model was established under some conditions. Numerical experiments on simulated and real datasets show the effectiveness of the proposed 3DTCTV and 3DNP methods.

REFERENCES

[1] H. K. Aggarwal and A. Majumdar, "Hyperspectral image denoising using spatio-spectral total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 442–446, Mar. 2016.

[2] M. Ahmad et al., "Hyperspectral image classification–traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[3] X. Cao, X. Fu, C. Xu, and D. Meng, "Deep spatial–spectral global reasoning network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504714.

[4] Y. Chang, L. Yan, X.-L. Zhao, H. Fang, Z. Zhang, and S. Zhong, "Weighted low-rank tensor recovery for hyperspectral image restoration," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 4558–4572, Nov. 2020.

[5] Y. Chen, Y. Guo, Y. Wang, D. Wang, C. Peng, and G. He, "Denoising of hyperspectral images using nonconvex low rank matrix approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5366–5380, Sep. 2017.

[6] Y. Chen, W. He, N. Yokoya, and T.-Z. Huang, "Hyperspectral image restoration using weighted group sparsity-regularized low-rank tensor decomposition," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3556–3570, Aug. 2020.

[7] Y. Chen, W. He, N. Yokoya, T.-Z. Huang, and X.-L. Zhao, "Nonlocal tensor-ring decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1348–1362, Feb. 2020.

[8] Y. Chen, T.-Z. Huang, W. He, X.-L. Zhao, H. Zhang, and J. Zeng, "Hyperspectral image denoising using factor group sparsity-regularized nonconvex low-rank approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515916.

[9] Y. Chen, F. Yuan, W. Lai, J. Zeng, W. He, and Q. Huang, "Low-rank tensor meets deep prior: Coupling model-driven and data-driven methods for hyperspectral image reconstruction," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 11, pp. 11685–11697, Nov. 2025.

[10] Y. Chen, H. Zhang, Y. Wang, Y. Yang, and J. Wu, "Flex-DLD: Deep low-rank decomposition model with flexible priors for hyperspectral image denoising and restoration," *IEEE Trans. Image Process.*, vol. 33, pp. 1211–1226, 2024.

[11] P. Coupe, P. Yger, S. Prima, P. Hellier, C. Kervrann, and C. Barillot, "An optimized blockwise nonlocal means denoising filter for 3-D magnetic resonance images," *IEEE Trans. Med. Imag.*, vol. 27, no. 4, pp. 425–441, Apr. 2008.

[12] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.

[13] H. Fan, Y. Chen, Y. Guo, H. Zhang, and G. Kuang, "Hyperspectral image restoration using low-rank tensor recovery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 10, pp. 4589–4604, Oct. 2017.

[14] H. Fan, C. Li, Y. Guo, G. Kuang, and J. Ma, "Spatial–spectral total variation regularized low-rank tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6196–6213, Oct. 2018.

[15] J. Fan and R. Li, "Variable selection via nonconcave penalized likelihood and its Oracle properties," *J. Amer. Stat. Assoc.*, vol. 96, no. 456, pp. 1348–1360, Dec. 2001.

[16] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.

[17] X. Guo, X. Huang, L. Zhang, and L. Zhang, "Hyperspectral image noise reduction based on rank-1 tensor decomposition," *ISPRS J. Photogramm. Remote Sens.*, vol. 83, pp. 50–63, Sep. 2013.

[18] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[19] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[20] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.

[21] C. J. Hillar and L.-H. Lim, "Most tensor problems are NP-hard," *J. ACM*, vol. 60, no. 6, pp. 1–39, Nov. 2013.

[22] F. L. Hitchcock, "The expression of a tensor or a polyadic as a sum of products," *J. Math. Phys.*, vol. 6, nos. 1–4, pp. 164–189, Apr. 1927.

[23] K. Huang, W. Kong, M. Zhou, W. Qin, F. Zhang, and J. Wang, "Enhanced low-rank tensor recovery fusing reweighted tensor correlated total variation regularization for image denoising," *J. Sci. Comput.*, vol. 99, no. 3, p. 69, Jun. 2024.

[24] M. E. Kilmer and C. D. Martin, "Factorization strategies for third-order tensors," *Linear Algebra its Appl.*, vol. 435, no. 3, pp. 641–658, Aug. 2011.

[25] D. Li, D. Chu, X. Guan, W. He, and H. Shen, "Adaptive regularized low-rank tensor decomposition for hyperspectral image denoising and destriping," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513917.

[26] Y. Li, J. Li, J. He, X. Liu, and Q. Yuan, "An optimization-driven network with knowledge prior injection for HSI denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531717.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:57:01 UTC from IEEE Xplore. Restrictions apply.

[27] B. Lin, X. Tao, and J. Lu, "Hyperspectral image denoising via matrix factorization and deep prior regularization," *IEEE Trans. Image Process.*, vol. 29, pp. 565–578, 2020.

[28] P. Liu, H. Long, Z. Zheng, N. Huang, and L. Xiao, "Gradient subspace-regularized hyperspectral image and stripe-coupled nonconvex tensor low-rank priors for destriping and denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5507218.

[29] P. Liu, J. Peng, H. Wang, D. Hong, and X. Cao, "Infrared small target detection via joint low rankness and local smoothness prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5708315.

[30] W. Liu and J. Lee, "A 3-D atrous convolution neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5701–5715, Aug. 2019.

[31] Y.-Y. Liu, X.-L. Zhao, Y.-B. Zheng, T.-H. Ma, and H. Zhang, "Hyperspectral image restoration by tensor fibered rank constrained optimization and plug-and-play regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500717.

[32] Y.-S. Luo, X.-L. Zhao, T.-X. Jiang, Y.-B. Zheng, and Y. Chang, "Hyperspectral mixed noise removal via spatial–spectral constrained unsupervised deep image prior," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9435–9449, 2021.

[33] A. Maffei, J. M. Haut, M. E. Paoletti, J. Plaza, L. Bruzzone, and A. Plaza, "A single model CNN for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2516–2529, Apr. 2020.

[34] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.

[35] I. V. Oseledets, "Tensor-train decomposition," *SIAM J. Sci. Comput.*, vol. 33, no. 5, pp. 2295–2317, Jan. 2011.

[36] H. Othman and S.-E. Qian, "Noise reduction of hyperspectral imagery using hybrid spatial–spectral derivative-domain wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 2, pp. 397–408, Feb. 2006.

[37] J. Peng, H. Wang, X. Cao, X. Liu, X. Rui, and D. Meng, "Fast noise removal in hyperspectral images via representative coefficient total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546017.

[38] J. Peng et al., "Learnable representative coefficient image denoiser for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506516.

[39] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[40] D. Qiu, M. Bai, M. K. Ng, and X. Zhang, "Nonlocal robust tensor recovery with nonconvex regularization *," *Inverse Problems*, vol. 37, no. 3, Mar. 2021, Art. no. 035001.

[41] D. Qiu, S. Xia, B. Yang, B. Li, and X. Zhang, "Poisson tensor completion via nonconvex regularization and nonlocal self-similarity for multi-dimensional image recovery," *J. Sci. Comput.*, vol. 102, no. 3, p. 76, Mar. 2025.

[42] D. Qiu, B. Yang, and X. Zhang, "Robust tensor completion via dictionary learning and generalized nonconvex regularization for visual data recovery," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11026–11039, Nov. 2024.

[43] K. Shi, J. Peng, J. Gao, Y. Luo, and S. Xu, "Hyperspectral image denoising via double subspace deep prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531015.

[44] Q. Shi, X. Tang, T. Yang, R. Liu, and L. Zhang, "Hyperspectral image denoising using a 3-D attention denoising network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10348–10363, Dec. 2021.

[45] G. Song, M. K. Ng, and X. Zhang, "Robust tensor completion using transformed tensor singular value decomposition," *Numer. Linear Algebra Appl.*, vol. 27, no. 3, pp. e229–9, May 2020.

[46] G.-J. Song, M. K. Ng, and X. Zhang, "Tensor completion by multi-rank via unitary transformation," *Appl. Comput. Harmon. Anal.*, vol. 65, pp. 348–373, Jul. 2023.

[47] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[48] X. Su, Z. Zhang, and F. Yang, "Fast hyperspectral image denoising and destriping method based on graph Laplacian regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511214.

[49] Y. Su, H. Zhu, K.-C. Wong, Y. Chang, and X. Li, "Hyperspectral image denoising via weighted multidirectional low-rank tensor recovery," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 2753–2766, May 2023.

[50] X. Tian, K. Xie, and H. Zhang, "A low-rank tensor decomposition model with factors prior and total variation for impulsive noise removal," *IEEE Trans. Image Process.*, vol. 31, pp. 4776–4789, 2022.

[51] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, Sep. 1966.

[52] H. Wang, J. Peng, W. Qin, J. Wang, and D. Meng, "Guaranteed tensor recovery fused low-rankness and smoothness," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10990–11007, Sep. 2023.

[53] M. Wang et al., "Tensor decompositions for hyperspectral data processing in remote sensing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 1, pp. 26–72, Mar. 2023.

[54] Y. Wang, Y. Han, K. Wang, and X.-L. Zhao, "Total variation regularized nonlocal low-rank tensor train for spectral compressive imaging," *Signal Process.*, vol. 195, Jun. 2022, Art. no. 108464.

[55] Y. Wang, J. Peng, Q. Zhao, Y. Leung, X.-L. Zhao, and D. Meng, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.

[56] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[57] S. Xia, M. K. Ng, and X. Zhang, "Low-rank tensor learning by generalized nonconvex regularization," 2024, *arXiv:2410.18402*.

[58] F. Xiong, J. Zhou, S. Tao, J. Lu, J. Zhou, and Y. Qian, "SMDS-Net: Model guided spectral–spatial network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 5469–5483, 2022.

[59] S. Xu, Q. Ke, J. Peng, X. Cao, and Z. Zhao, "Pan-denoising: Guided hyperspectral image denoising via weighted represent coefficient total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528714.

[60] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[61] J. Xue, Y. Zhao, W. Liao, and S. G. Kong, "Joint spatial and spectral low-rank regularization for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1940–1958, Apr. 2018.

[62] J. Xue, Y.-Q. Zhao, T. Wu, and J. C.-W. Chan, "Tensor convolution-like low-rank dictionary for high-dimensional image representation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13257–13270, Dec. 2024.

[63] Y. Yang, J. Zheng, S. Chen, and M. Zhang, "Hyperspectral image restoration via local low-rank matrix recovery and moreau-enhanced total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1037–1041, Jun. 2020.

[64] H. Ye, H. Li, B. Yang, F. Cao, and Y. Tang, "A novel rank approximation method for mixture noise removal of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4457–4469, Jul. 2019.

[65] J. Ye, F. Xiong, J. Zhou, and Y. Qian, "Iterative low-rank network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528015.

[66] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[67] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[68] C.-H. Zhang, "Nearly unbiased variable selection under minimax concave penalty," *Ann. Statist.*, vol. 38, no. 2, pp. 894–942, Apr. 2010.

[69] H. Zhang, H. Chen, G. Yang, and L. Zhang, "LR-Net: Low-rank spatial–spectral network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 30, pp. 8743–8758, 2021.

[70] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.

[71] H. Zhang, T.-Z. Huang, X.-L. Zhao, W. He, J. K. Choi, and Y.-B. Zheng, "Hyperspectral image denoising: Reconciling sparse and low-tensor-ring-rank priors in the transformed domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502313.

[72] H. Zhang, L. Liu, W. He, and L. Zhang, "Hyperspectral image denoising with total variation regularization and nonlocal low-rank tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3071–3084, May 2020.

[73] Q. Zhang, Q. Yuan, M. Song, H. Yu, and L. Zhang, "Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 6356–6368, 2022.

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[74] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, and Y. Xiao, "Hyperspectral image denoising: From model-driven, data-driven, to model-data-driven," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13143–13163, Oct. 2024.
[75] X. Zhang and M. K. Ng, "Low rank tensor completion with Poisson observations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4239–4251, Aug. 2022.
[76] Q. Zhao, G. Zhou, S. Xie, L. Zhang, and A. Cichocki, "Tensor ring decomposition," 2016, *arXiv:1606.05535*.
[77] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, T.-H. Ma, and T.-Y. Ji, "Mixed noise removal in hyperspectral image via low-fibered-rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 734–749, Jan. 2020.
[78] L. Zhuang and M. K. Ng, "FastHyMix: Fast and parameter-free hyperspectral image mixed noise removal," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4702–4716, Aug. 2023.
[79] L. Zhuang, M. K. Ng, L. Gao, and Z. Wang, "Eigen-CNN: Eigenimages plus eigennoise level maps guided network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512018.



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